Ms. Saba Ozhar

SEAT NO. CT- 005

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2021

Time: 3 Hours

Dated:10-02-2023
Max.Marks:60

System Analysis & Design - CT-259

Q1) Look at the Figure of Cost-Benefit Analysis below to answer the following questions. [CLO-3, 12 marks]

	2003	2004	2005	2006	2007	Total
2 Servers @ $125,000	250,000	0	0	0	0	
Printer	100,000	0	0	0	0	
Software licenses	34,825	0	0	0	0	
Server software	10,945	0	0	0	0	
Development labour	1,236,525	0	0	0	0	
Total Development Costs:	1,632,295	0	0	0	0	
Hardware	54,000	81,261	81,261	81,261	81,261	
Software	20,000	20,000	20,000	20,000	20,000	
Operational labour	111,788	116,260	120,910	125,746	130,776	
Total Operational Costs:	185,788	217,521	222,171	227,007	232,037	
Total Costs:	1,818,083	217,521	222,171	227,007	232,037	
PV of Costs	1,765,129	205,034	203,318	201,693	200,157	2,575,331
PV of All Costs:	1,765,129	1,970,163	2,173,481	2,375,174	2,575,331	
Tangible Benefits						
Increased sales	500,000	530,000	561,800	595,508	631,238	
Reduction in customer complaint calls	70,000	70,000	70,000	70,000	70,000	
Reduced inventory costs	68,000	68,000	68,000	68,000	68,000	
Total Benefits:	638,000	668,000	699,800	733,508	769,238	
PV of Benefits:	619,417	629,654	640,416	651,712	663,552	3,204,751
PV of All Benefits	619,417	1,249,071	1,889,487	2,541,199	3,204,751	
Total Project Costs Less Benefits:	1,180,083	450,479	477,629	506,501	537,201	
Yearly NPV:	1,145,712	424,620	437,098	450,019	463,395	629,420
Cumulative NPV:	-1,145,712	-721,091	-283,993	166,026	629,421	

a) **Calculate** the following: (i) Return on Investment (2)
(ii) Break-even Point (2)

b) **Draw** the graph of the break-even point (4)

c) **Ascertain**, giving examples, what development costs, operational costs, tangible benefits and intangible benefits are (4)

Q2) **Elaborate** briefly what the six Information gathering techniques are [CLO-2, 12 marks] (12)

Q3) **Identify** Functional and Non-functional requirements [CLO-1, 12 marks] (12)

P.T.O

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Q4) The Event table below shows events in the order-entry subsystem. Use the Event table to answer the following questions. [CLO-3, 12 marks]

Entities in ERD: Catalog Product, Catalog, Product item, Inventory item, Shipper, Return item, Order item, Shipment, Customer, Order, Order Transaction

Order-entry subsystem event table					
Event	Trigger	Source	Use case	Response	Destination
1. Customer wants to check item availability	Item inquiry	Customer	Look up item availability	Item availability details	Customer
2. Customer places an order	New order	Customer	Create new order	Real-time link Order confirmation Order details Transaction	Credit bureau Customer Shipping Bank
3. Customer changes or cancels order	Order change request	Customer	Update order	Change confirmation Order change details Transaction	Customer Shipping Bank
4. Time to produce order summary reports	"End of week, month, quarter and year"		Produce order summary repots	Order summary reports	Management
5. Time to produce transaction summary reports	"End of day"		Produce transaction summary reports	Transaction summary reports	Accounting

a) **Draw** five separate DFD fragments for the order-entry subsystem (5)
b) **Draw** the event-partitioned model for the order-entry subsystem (4)
c) **Draw** the detailed DFD for the 'Customer places an order' DFD fragment (3)

Q5) Answer the following questions: [CLO-3, 12 marks]

a) **Draw** a use case diagram for the Order-entry subsystem. Use the <<includes>> relationship (4)
b) **Draw** the Activity diagram for the Telephone Order Scenario. Include the three swimlanes 'Customer', 'Order Clerk' and 'Computer System' (4)
c) **Draw** the resulting System Sequence Diagram (SSD) for the Telephone Order Scenario (4)

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SEAT NO. CT-20032

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN
COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
FALL SEMESTER EXAMINATIONS 2021
BATCH 2020

Time: 3 Hours

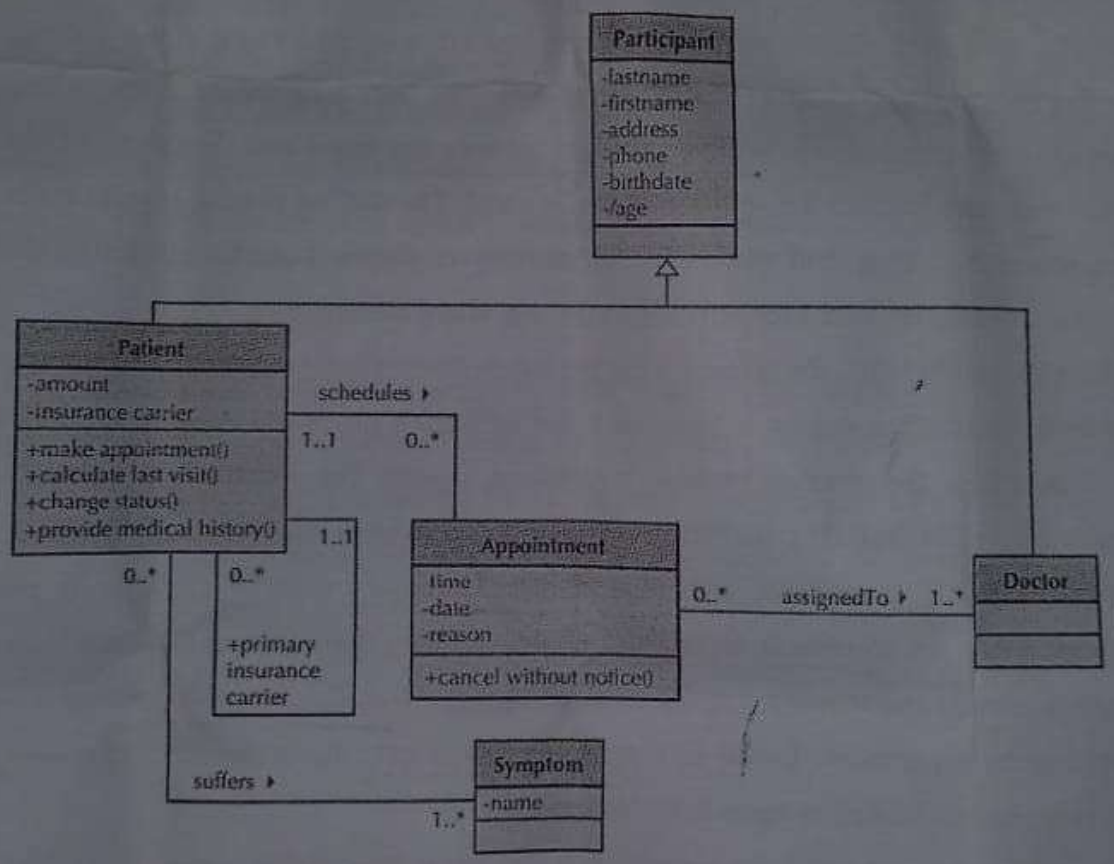
Dated:19-02-2022
Max.Marks:60

System Analysis & Design - CT-259

Instructions:
Attempt all questions.
Avoid unnecessary details.
Any assumptions must be mentioned clearly

Question#1 **(12 marks)**

CRC (Class–Responsibility–Collaboration) cards should be initiated after reviewing use case scenarios but in this question consider the following class diagram of "Appointment Process" and **CREATE** CRC cards of 'Participant', 'Patient' and 'Appointment'. **HOW** Role-play scenarios help in developing CRC cards.



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CASE STUDY FOR QUESTION#2 and QUESTION#3

Influencia is a management company whose objective is to hire highly skilled technical employees on a short-term, temporary basis to assist with special projects or to provide needed technical skills to the client companies. This company (Influencia) negotiates contracts with its client company in which it agrees to provide temporary staff in specific job categories for a specified cost. For instance, Influencia has a contract with an oil and gas exploration company in which it agrees to supply geologists with at least a master's degree for $5,000 per week. Influencia has contracts with a wide range of companies and can place almost any type of professional or scientific staff members, from computer programmers to geologists to astrophysicists.

When any client (requesting company) determines that it will need a temporary professional or scientific employee, it issues a staffing request against the contract it had previously negotiated with Influencia. When Influencia's contract manager receives a staffing request, the contract number referenced on the staffing request is entered into the contract database. Using information from the database, the contract manager reviews the terms and conditions of the contract and determines whether the staffing request is valid. The staffing request is valid if the contract has not expired, the type of professional or scientific employee requested is listed on the original contract, and the requested fee falls within the negotiated fee range.

If the staffing request is valid, the contract manager enters the staffing request into the staffing request database. The staffing request is then sent to the Influencia placement department. In the placement department, the staffing request is checked against the database of available professional and scientific staff. If a qualified individual is found, he or she is marked "reserved" in the staff database. If a qualified individual cannot be found in the database or is not immediately available, the placement department creates a memo that explains the inability to meet the staffing request and attaches it to the staffing request. All staffing requests are then sent to the arrangements department. In the arrangements department, the prospective temporary employee is contacted and asked to agree to the placement.

If the staffing request is not valid, the contract manager sends the staffing request back to the client with a justification letter.

Question #2 **(12 marks)**

CREATE a use-case diagram for the system described here. **WRITE** use case description of any two use cases. You may add assumption to complete this case.

Question#3 **(12 marks)**

Use the use case descriptions that you have developed for Question#2 and **CREATE** two activity diagrams for the selected use cases. **HOW** will you verify and validate your developed activity diagrams?

Question#4 **(12 marks)**

Consider the following case; The University has several departments. Each department is managed by a chair, and at least one professor. Professors must be assigned to one, but possibly more departments. At least one professor teaches each course, but a professor may be on leave and not teach course. Each course may be taught more than once by different professors. We know of the department name, the professor name, the professor employee id, the course names, the course schedule, the term/year that the course is taught, the departments the professor is assigned to, the department that offers the course. **DRAW** an Entity Relationship diagram (ERD) for this situation and place minimum and maximum cardinalities on the diagram. Add assumptions if required. **GIVE** your own example of repeating groups in an un-normalized design (You may use the same case for creating your example).

Question#5 **(12 marks)**

User Experience is important to research for creating User Interface. Google Class Room has been used by majority of users during Covid-19. **DISCUSS** features of Google Class Room as per your experience as a student by considering User Interface Design principles. **DRAW** story board for including any new feature for students to enhance the use of Google Class Room. **(Only create story board of new feature).**

M. Abdul Rafay.

Roll#: CT-12 **Fall Semester 2021 SS (CS) Sec-A MID TERM** **Date: 13-12-2021**

Total Marks: 20

Time : 1 hour 40 minutes

Important Instructions:

1. Only A-4 size paper will be evaluated. No other size of the paper will be considered.
2. Solve your paper using pen. You may use pencils for diagram but use pens for all naming conventions.

Question#1: a) DRAW PERT chart for the given Work breakdown structure to estimate project duration. Also highlight the critical path in your diagram. Use the timelines according to the following instructions;

(8 marks)

Odd Roll numbers should use "Most likely time".

Even Roll numbers should use "Optimistic time".

B

Activity	Optimistic Time	Most Likely Time	Pessimistic Time	
A	3	7	11	B
B	5	9	13	—
C	1	2	9	A
D	2	3	16	C
E	2	4	18	A
F	3	4	11	D
G	1	4	7	H
H	3	4	5	A
I	2	4	12	F, G, J
J	4	7	9	H, E

OR

Question#1: a)

Consider the following case; Make any assumptions to complete this case.

A hospital has a large number of registered physicians. Attributes of Physician include physicianID and specialty. Patients are admitted to the hospital by physicians. Attributes of Patient include patientID and patientName. Any patient who is admitted must have exactly one admitting physician. A physician may optionally admit any number of patients. Once admitted, a given patient must be treated by at least one physician. A particular physician may treat any number of patients, or may not treat any patients. Whenever a patient is treated by physician, a hospital wishes to record the treatment details of the patient, by including date, time and result of the treatment. **DRAW** a class diagram of the given case with complete notations and cardinalities. You may add your realistic assumptions to complete this case.

Question#1: b) "Selection of tools that are used in the development of system analysis and design must be based on providing user friendly interface". Consider any example tool to justify this statement.

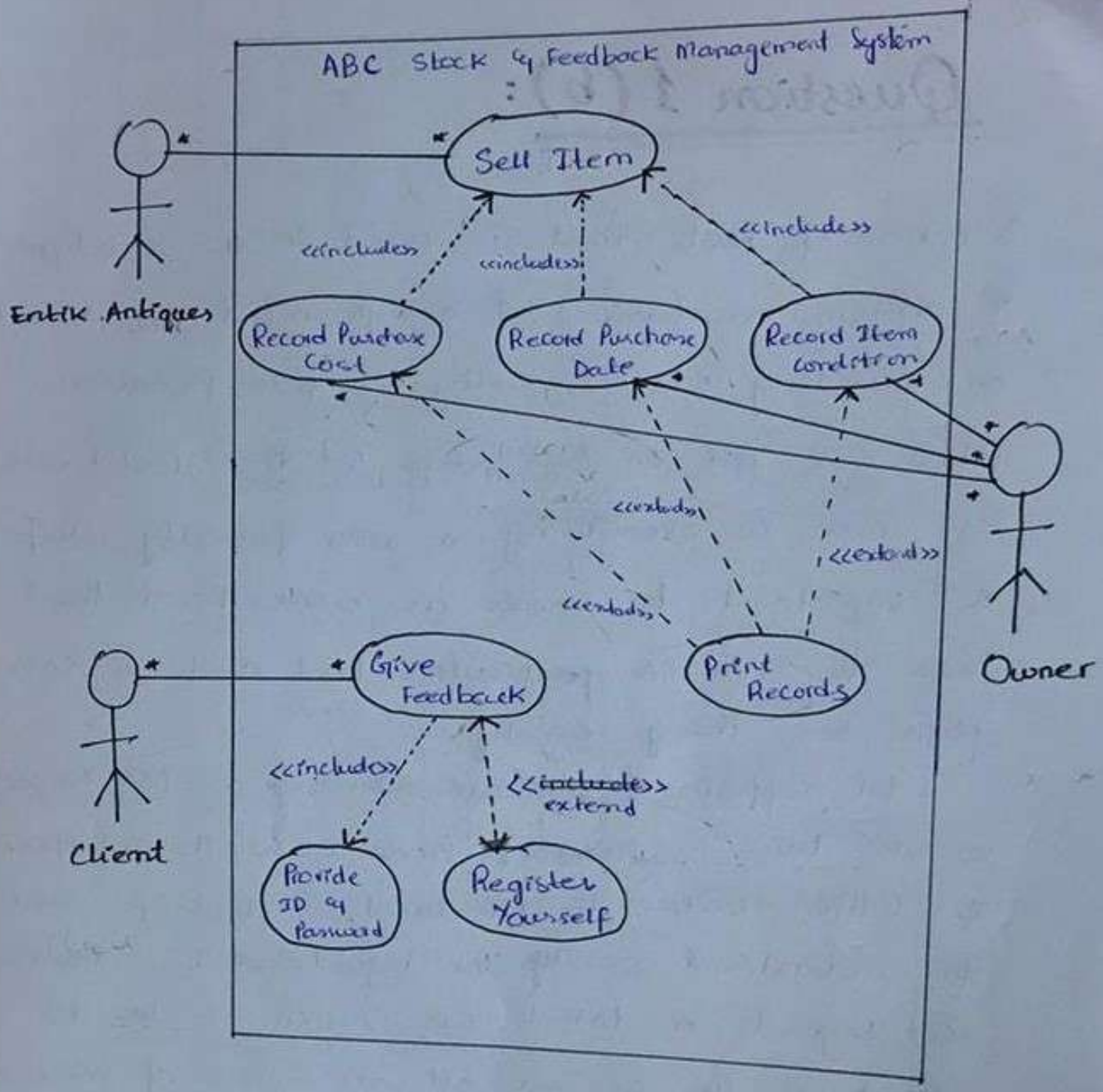
(2 marks)

Question#2: a) Entik Antiques buys and sells one of a kind antiques of all kinds (for example, furniture, jewellery and clothing). Entik Antiques works with many different individuals, called clients, who sell items to and buy items from the store. Some clients only sell items to Entik Antiques, some only buy items, and some both sell and buy. When the Entik Antiques sells an item in stock to the client, the owners want to record the purchase cost, date purchased and condition at the time of purchase. After one month, Entik Antiques want their clients to give feedback for their service. Only registered clients can give feedback to Entik Antiques by providing their valid identity (ClientID and password). **DRAW** a use-case diagram for the given case. Also, **MAKE** description tables for any two use-cases. You may add your realistic assumptions to complete this case.

Question#2: b) Identify any mistakes/incomplete details in the given user story. Justify your identified points. (2 marks)

Story card # 1
Title: Customer
As a Customer,
I can buy items and pay bills.

Question 2(a):



Assumption: I have assumed that the owner may be used to print the records or may be not.

SEAT NO. CT-18028

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR FALL SEMESTER (BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY)
EXAMINATIONS 2019
BATCH 2018

Time: 3 Hours | **Dated:08-02-2020**
Max.Marks:60

System Analysis & Design - CT-259

Instructions:

- Attempt all Questions
- Draw neat and clean diagrams
- Avoid unnecessary details

Question#1 **(12 marks)**

CREATE a context level and Level 0 DFD of a Meal-Delivery service system. A system must store information about Restaurants, Menu Items, Customers, Orders, Order payments and Drivers. Adding details about these processes, a system can also create/add new restaurant, a new driver can be hired and details can be saved in system. There are main two modules first is Admin and second is User or Customer. Admin is a responsible person who operate the whole system. Admin has rights to add, delete, modify category, item and all other information regarding to meal-delivery system. User is a visitor of our website, who visit the site and select some order for food and can make call to order. Payment can be made using cash on delivery option. DRAW level 1 diagram of any one process. State your assumptions clearly. more

Question#2 **(12 marks)**

DRAW an activity diagram for the following employee hiring process. Projects, Inc., is an engineering firm with approximately 500 engineers in different specialties. New employees are hired by the personnel manager, based on data in an application form and evaluations collected from other managers who interview the job candidates. Prospective employees may apply at any time. Engineering managers notify the personnel manager when a job opens and list the characteristics necessary to be eligible for the job. The personnel manager compares the qualifications of the available pool of applicants with the characteristics of an open job and then schedules interviews between the manager in charge of the open position and the three best candidates from the pool. After receiving evaluations on each interview from the manager, the personnel manager makes the hiring decision based upon the evaluations and applications of the candidates and the characteristics of the job, and then notifies the interviewees and the manager about the decision. Applications of rejected applicants are retained for one year, after which time the application is purged. When hired, a new engineer completes a nondisclosure agreement, which is filed with other information about the employee.

P.T

- Quality Engineers (perform manual testing of Electron)
- Automation Engineers (write test automation suits for Electron)
- Performance Engineers (perform performance testing of Electron)
- Security Engineers (find security loop holes in Electron and fix them)
- Assistant Manager Marketing (responsible for marketing and promotions of Electron)
- Assistant Manager HR (responsible for employee policies and recruitment)

Apply best class hierarchy for this scenario. Add necessary details to it.

Question.4 (12 marks)

What is containership and delegation? Consider a simple Shopping Cart where users have options to pay for their purchases in two different ways – using Credit Card or using PayPal. Extend the class diagram by making appropriate connection of PayPal and credit card with Bank account. Add necessary detail to complete this case and apply the implementation of best design pattern(s).

Question.5 (12 marks)

Create a class by name measure with the member data feet and inches. Perform the following:

a) Overload >= operator.
b) Overload == operator.
c) Overload + operator to add two measurements.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR FALL SEMESTER (BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY)
EXAMINATIONS 2018
BATCH 2017

Dated:14-02-2019
Max.Marks:60

Time: 3 Hours

System Analysis & Design - CT-259

Instructions:

- **Attempt all Questions.**
- **All Questions carry equal marks.**
- **Draw diagrams where required and avoid unnecessary details.**

Question#1:

LIST and EXPLAIN some of the problems with the traditional waterfall SDLC. EXPLAIN manifesto of adopting Agile Methodology. **(12 marks)**

Question#2:

DRAW a Network diagram for the activities mentioned in the table below. Use forward and backward pass to DETERMINE project duration and critical path. **(12 marks)**

Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3

Question#3:

WHAT is object-oriented analysis and design? Consider an online airline booking system where passengers can view flight schedule, select preferable seats and book tickets. Tickets must be booked by online payment using Credit Card. DRAW the Entity Relationship Diagram (ERD) and Class diagram by making appropriate connection of credit card with Bank account. (State your assumptions clearly to complete this case study) **(12 marks)**

$ES_i = Max(ES_i + D_i)$

$LC = Min(LC_i - D_i)$

P.T.O

Question.3 (12 marks)

Consider a company that is in the business of renting DVDs and video tapes. The company has recently entered into e-business and the senior management wants to make the existing system to Web-enabled system. DESIGN and IMPLEMENT a system that manage user login information and allows users to pay rent of DVDs and video tapes online via credit card or debit card. Users may also be allowed to go for an option of "cash on delivery". Add necessary details to this case-study as per your design.

Question.4 (12 marks)

Universities are simultaneously a real estate property management company (residential student housing), restaurant with multiple outlets (dining halls), retailer (bookstore), events management and ticketing agency (athletics and speaker events), police department (campus security), professional fundraiser (alumni development), consumer financial services company (financial aid), investment firm (endowment management), venture capitalist (research and development), job placement firm (career planning), construction company (buildings and facilities maintenance), and medical services provider (health clinic). In addition to these varied functions, higher education institutions are obviously also focused on attracting high caliber students and talented faculty to create a robust educational environment. Your responsibility is to select any 'ONE' of the mentioned systems, add necessary functionalities to your design and APPLY appropriate design to implement the selected system.

Question.5 (12 marks)

"Prefer composition over inheritance". GIVE example case study that APPLY concepts to prove this statement. Example case and its design should include complete details of your example.

Restaurants

a) ~~details~~ → billing name, no. of people

b) ~~discount~~:

1) Desi → dal chawal, biryani

2) Pan Asian → chinese, chowmein

3) Italian → Pasta, pizza

4) ~~American~~ Fast Food → burger, roll, fries

5) Desserts → choc cake, icecream

c) Bill → billing name, selected food items, price

venture capitalist startups/Research:

1) Engineering

2) IT

3) Medical

a) Research/startup name

b) description, targeted audience

d) How will it help